Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 56.2 Friction Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · CLASS 56.2 Friction Numerical
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

CLASS 56.2 Friction Numerical

1 CLASS 56.2 Friction Numerical

Study focus: CLASS 56.2 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000174 · EV0000175

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Problem six slash three.

Key details

- The force peace applied to the 50 kg block when it is at rest.
- Determine the magnitude and direction of the friction force exerted by the surface on the block if the.
- P is equals to zero, second P is equals to two hundred newton, third case when P is to fifty newton, fourth, when what value of P is required to.
- Initiate motion up the incline.
- The coefficient of static and the kinetic friction between the black and the inclined mu is equals to 0.25 and mu case equals to 0.2 respectively.
- Then there is a fifty kg crate resting on this inclined plane.
- It is in a state of rest.
- Then we are going to apply this force P.
- Due to this force, this block will have a tendency to move up the.
- Then in the last they have asked us for what value of this P this block will start to move up the inclined plane.
- That's when p is equals to zero.
- Next, the weight of this block will be vertically downward.
- Then the components of this W parallel x axis will be up the incline sored plane and the component of W.
- Parallel to y axis will be downward like this.
- Then P, the components of P will be right and upward.
- Since here it is P is 0, you can neglect.

- This for time being, then there will be a normal reaction from the surface onto the plane that will be upward.
- Since the value of P is 0 this block will have a tendency to slide down the plane.
- This is the complete free body diagram of the block.
- Since n is upward, it will it will be plus n and the y component of w is downwards, so minus w into sine so cos 15 degree because the angle 15 is with respect to.
- Why the axis so cost 15 degree.
- Now apply the second condition of equilibrium sigma if x is equal to 0.
- Then the x component of this w is down the plane, so that will be minus w into sine fifteen degree.
- From this we can find the value of F.
- F is 126.95 newton.
- That we will get to know.
- Only when we compare this frictional force with the maximum frictional value.
- We know f max is equals to mu s into n, here mu s is 0.25 and n we know from here it is 475.78.
- Simplify we are going to get the value of f maxes under an 18.44 newton.
- Since F is greater than F max, this condition is not at all possible.
- It is impossible.
- If the block is moving down the plane, then the friction between the block and the surface will be.

Example or application from the lecture

- Here what is given is there is an inclined plane.
- This inclination of the plane with the horizontal is given as fifteen degree.
- And they have given the coefficient of static friction as 0.25 and kinetic friction as point two.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 05:00–09:59

Kinetic friction that will be equals to mu c and two n.

Key details

- Here mu k is point two and n we know four seventy three point seven eight.
- So actual friction between the block and the plane when p is equals to zero is seventy four point seven five newton and that is up the plane because the block is moving down the plane.
- Next, the second case when the value of P is two hundred newton.
- Like this, x axis and the y axis in this direction, normal to the x axis, then the cell fate vertically downward, then the components of W will be like this, then components of P going to be P cos 20 and P sin 20.
- Normal reaction up the plane and the frictional force will be.

- Since the value of P is 200 newton we are going to assume because of this P this block will have a tendency to move up the plane, so the frictional force will be down the plane.
- That's why I have considered the friction in this direction.
- Then again the same conditions of equilibrium, sigma f of y and sigma f of x is equal to 0.
- N is upwards to plus n.
- This P sin 20 is also upwards, so plus P sin 20, W is downward, so minus W into cos 15 degree will be equal to 0.
- Here only one unknown that is N.
- Substitute the value of P s 200, we'll get the value of NS 542.2 newton.
- Next sigma Fx is equal to 0.
- X component of W is towards left, so minus W into sin twelve fifteen will be equals to zero.
- Here also only one unknown that is F.
- Further what we can do here is we have to compare this.
- Value of F with respect to F max.
- We know that F max is equal to mu S into N.
- N muus is point two five and n is five forty two point two.
- Find the value of F max.
- We'll get F max as 135.55 newton.
- With respect to F max, since F is smaller than F max, this indicates the body is in a state of rest.
- Once the body is under rest, the frictional force will be this itself, that is.
- Sixty point nine N and since the answer is positive that indicates our assumption is correct.
- We assumed the block is having a tendency to move up the plane, so frictional forces down the plane.
- This is correct.
- The third condition.
- When P is two fifty N, the same wavy body diagram.
- Since the value of p is two fifty newton we will assume the block is moving up the plane so fictional force will be down the plane.
- Apply the conditions of equilibrium sigma f y 0.
- N is upward, so plus n, then p sin twenty is also upward, so plus p sin 20, then the y component of w is downwards, so minus w.
- Cause fifteen will be equal to zero.
- Here also only un one unknown that is n.
- Next second condition of equilibrium, sigma f of x is equal to zero.
- f is towards left, so minus f.
- P component is towards right.
- Here also only one unknown that is CF.
- We'll get the value of F as 107.97 N.
- That we will get to know only when we compare the value of this F with respect to F max.
- N is double five nine point three and s is point two five we'll get the value of f max is one thirty five point five newton.
- Since f is smaller than f max this indicates the body is under rest.
- So therefore f will be equals to.

Example or application from the lecture

- It is the same free body diagram normal to the consider x axis along the plane, but.
- Let's start with sigma f y 0.
- Once we consider sigma fx is equal to 0, f is left side, so minus f, x component of p is towards right, so plus p cos twenty.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 10:00–14:59

107.95 newton and we assume the block is having a tendency to move up the plane so fictional forces down the plane.

Key details

- And we got the answer also as positive.
- This is the last case for what value of P this fifty kg block will start to move up the incline.
- That's what we have to calculate.
- The solution remains the same.
- Again we have to start with a free body diagram.
- Components of W are going to be like this.
- Then the components of P are going to be right side and upward, then up.
- It's going to be P cos twenty and this is going to be P sin twenty.
- Normal reaction will be upward.
- Since the block is moving up the plane, the frictional force will be down the plane.
- Next, start with the conditions of equilibrium.
- Sigma F of Y is equals to 0.
- N is upwards, so plus N.
- P component of P is upwards, so plus P sin 20, component of W is downwards, so minus W.
- Cos 15 will be equals to 0.
- Here we have two unknowns, one is P and one more is N.
- Call it as equation one.
- Now apply the second condition of equilibrium that is sigma fx is equal to 0.
- Since the component of F is left side, so minus F, it is right side, so plus P e c cost twenty, it is left side, so minus W sine fifteen will be equal to zero.
- Here also we have two unknowns F and P.
- Since the block is under motion, we can replace F as mu K into N because here the frictional force will be kinetic friction that will be always equal to mu K into N.
- Once we solve equation one and two, we are going to get the value of P s two thirty nine point three newton.
- That is for the meaning.
- Next question, problem 6 slash 6.
- here as I said before always x axis should be taken along the inclined plane.
- The components of W are going to be X component is going to be down the plane and the Y component is going to be downward and the angle here, since this angle is 15 degree, the

angle between W and the Y-axis will also be 15 degree.
- Next there will be a normal reaction from the surface unto the drum that will be upward and before that.
- P is that so we have to resolve this P.
- To resolve the P we should know the angle.
- If I draw a line passing through this, which is parallel to X axis, the angle between the P and the line parallel to X axis will be fifteen degree because that is going to be thirty minus fifteen, that will be fifteen degree.
- The component is going to be P cos fifteen and one more will be P sine fifteen degree.
- The next is normal reaction of the plane and it is going to move in this direction, the drum is moving in this direction, so the frictional force will be up the incline plane.
- Next, we have to apply the conditions of equilibrium.
- If you think of sigma Fx and FY, we have two unknowns.
- P is also your unknown and n is also unknown.

Example or application from the lecture

- Consider X along the plane and the Y axis normally to the plane and the weight vertically downward.
- Here mu K is given as.
- And the weight since the mass is given as m, we have to consider the weight w vertically downward.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 15:00–19:59

If you think of sigma fx zero, f is a unknown, but we can express f as mu s into n, that is correct.

Key details

- But N is also an unknown.
- W cos fifteen, then P sin 15 will be equal to 0 because they're acting at the same point.
- Since P is clockwise, it is going to be negative.
- Minus P cos 15 into perpendicular distance will be R by 2.
- Then moment of F will be equal to this counterclockwise direction.
- Once we apply sigma fx 0 plus f minus w sine 15 plus p cos 15 will be equals to 0.
- As we know the value of f substitute here f is point four eight three times that of p minus w is m into g into sin fifteen plus p cos fifteen.
- Solve this we are going to get the value of P as 0.78.1786 times it of mg.
- Next apply the next condition of equilibrium sigma f y 0 n is upwards so plus n then p sin 15 is also upward so plus.
- Then W cos 15 is downwards or minus w will be equal to 0.
- We already know the value of P and P in terms of Mg, so substitute here W is also equal to Mg.
- We will get the value of NS 0.919 times of Mg.

- Next we should know the value of coefficient of static friction mu s it will be equal to f by n.
- F is 0.483 times the of P and N is 0.919 times the of EMG.
- We'll get the value of mu s as 0.094.
- This is the coefficient of static friction between the drum and the surface for which this will have a tendency to roll up the plane.
- That's it for today's class.

Example or application from the lecture

- So let's start with the moment condition.
- We can consider moment about the center of the drum as 0.
- Once we consider moment about O as 0, the moment due to W sine 15, then.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 56.2 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.2 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.2 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 56.2 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...

C Revision and retrieval practice